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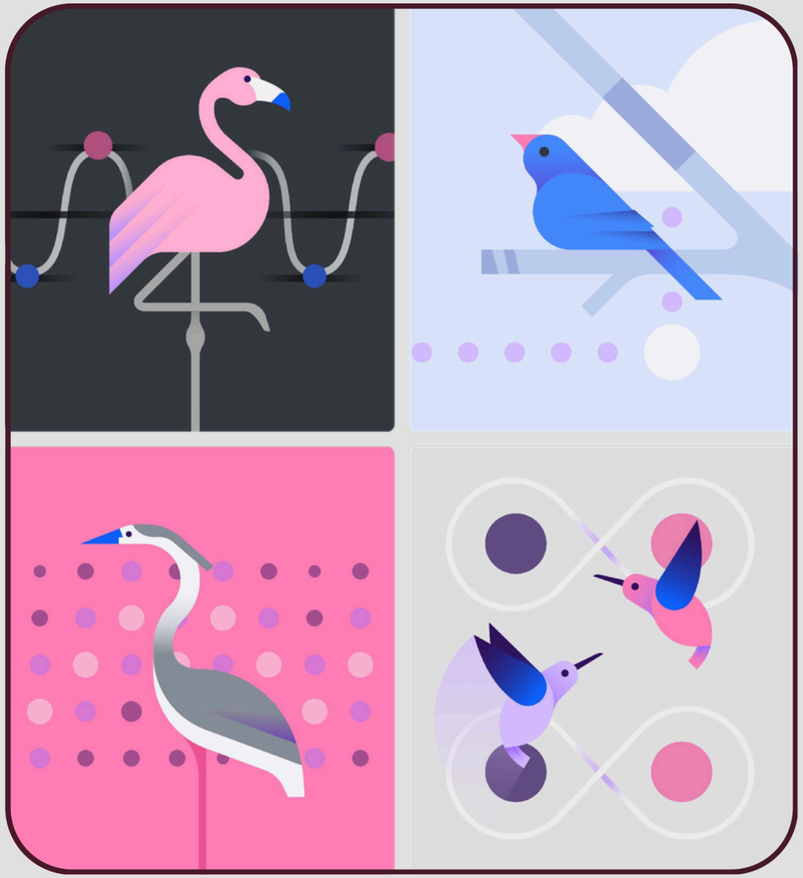
Q-HACK INDIA 2026

Team Name: BLOODFlow-Q

Team Lead Name : RAHUL ARYA AV

Track: Special Track — Open Quantum Innovation

Problem Statement title: BloodFlow-Q: Quantum Optimization for Emergency Blood Allocation



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Problem

During medical emergencies, blood resources become critically scarce while hospital demand fluctuates unpredictably. The core challenge is **dynamically allocating scarce**, **compatible blood units** across a distributed healthcare network while **balancing clinical urgency**, **strict biological compatibility,** and **transit constraints.**

NETWORK & SUPPLY BOTTLENECKS

- **Multi-Node Inventory Mismatch**: Multiple blood banks hold uneven, finite stocks across 8 blood groups serving competing hospital urgencies.
- **Systemic Failure of Local Heuristics**: Traditional "nearest-bank" greedy dispatching solves one hospital's need while triggering fatal shortages across the wider regional network.

OPERATIONAL CONSTRAINTS & SCALE

- **Rigid Multi-Factor Constraints**: Allocations must strictly satisfy ABO/Rh compatibility, physical inventory caps, golden-hour deadlines, and clinical triage preemption.
- **Combinatorial Explosion**: The assignment space grows exponentially, making real-time re-optimization impossible via manual phone calls during sudden mass-casualty surges.

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Why It Matters

The core issue is not simply whether blood exists in the system - it is whether **finite, perishable resources are allocated intelligently** across locations and clinical urgencies before time runs out.

Key Operational Question: "Given currently available compatible resources, where should they be allocated right now?"

WHO FACES THE PROBLEM

- **Emergency & Trauma Hospitals:** Requiring immediate access to compatible units during time-critical maternal and accident surgeries.
- **Distributed Blood Banks:** Managing finite inventories with severe shelf-life perishability (5-day platelets vs. 35-day PRBCs).
- **Regional Transfusion Services:** Coordinating movement across fragmented public, private, and NGO networks under road congestion.

WHY OPTIMIZATION IS ESSENTIAL

• **Triage Under Resource Deficits:** When 100 units exist for 150 requests, mathematical optimization guarantees critical demand is satisfied first.

- **Eliminating Avoidable Wastage:** Re-routing near-expiry units prevents the tragedy of hundreds of thousands of discarded units.
- **Balanced Network Resilience:** Preserves baseline safety reserves across peripheral clinics rather than hoarding units at city centres.

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Proposed Solution

BloodFlow-Q is a hybrid quantum-classical decision system that **dynamically reallocates scarce compatible blood units** across hospital networks under sudden demand shocks, strict biological compatibility, and transit constraints.

Inventory & Demand → LP Relaxation + QUBO → LP + Warm QAOA → Classical Validator → Optimal Dispatch

END-TO-END HYBRID ARCHITECTURE

- **Mathematical Problem Mapping:** Ingests regional blood bank stock, hospital urgency, and transit times, converting the allocation into a constrained QUBO / Ising Hamiltonian. **LP-guided quantum optimization:** Uses the continuous LP relaxation to bias the initial state, then executes parameterized Warm QAOA circuits on the local Qiskit Aer simulator to sample
- candidate allocations for classical validation.
  Rigorous Classical Post-Validation: Decodes output bitstrings and strictly verifies inventory conservation, ABO/Rh compatibility, and cold-chain constraints before dispatch.

EMERGENCY RE-OPTIMIZATION ENGINE

- **Dynamic Demand Shock Response:** When a highway mass-casualty incident or PPH crisis hits, the system reformulates and recomputes the regional schedule in sub-minute windows.
- **Active Triage & Preemption:** Dynamically releases non-urgent elective reserves to satisfy life-critical emergency trauma without creating secondary regional deficits.
- **Wastage & Perishability Mitigation:** Prioritizes near-expiry units (Day-4/5 platelets) for immediate clinical consumption, minimizing avoidable hospital discards.

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Hybrid Method

Emergency blood allocation is an **NP-hard constrained combinatorial problem**. Binary dispatch decisions and coupled non-linear interactions map natively to a **QUBO / Ising Hamiltonian**, with LP/MILP baselines and an LP-seeded Warm QAOA.



MATHEMATICAL MAPPING: WHY QUBO?

- **Binary Decision Space:** Formulates discrete allocation decisions x; Є {0, 1} to represent unit or clinical batch distribution.
- **Quadratic Constraint Penalties:** Converts hard operational constraints (mass conservation, supply caps) into squared penalty terms:
- Penalty = P • (x1 + x2 + x3 - Demand) ^2
- **Native Cross-Matching Coupling:** Biological incompatibility penalties and quadratic delay curves map directly to pairwise Ising interaction terms ($J_{ij}\ \sigma_i^z\ \sigma_j^z$).

LP-GUIDED WARM QAOA

- **Variational Ground-State Sampling:** Parameterized quantum circuits evolve the state using alternating Cost and Mixer Hamiltonians to explore combinatorial configurations.
- **Realistic Hybrid Co-Design:**
- *Classical CPU: SciPy HiGHS LP relaxation, QUBO*
- *mapping, COBYLA tuning, LP/MILP baselines and*
- *feasibility checks. Quantum simulator*: Warm QAOA sampling on Qiskit Aer; no quantum advantage or hardware result is claimed.

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Implementation and Architecture

BLOODFLOW-Q SYSTEM ARCHITECTURE

REACT DASHBOARD → FAST API BACKEND

CLASSICAL	QUANTUM
SCENARIO ENGINE	QUBO BUILDER
COMPATIBILITY ENGINE	ISING HAMILTONIAN
LP / MILP BASELINES	LP-WARM QAOA CIRCUIT
EMERGENCY-EVENT ENGINE	MEASUREMENT → BITSTRING
	ALLOCATION DECODER

VALIDATION ENGINE → RESULTS ENGINE → DASHBOARD

TECH: React | FastAPI | SciPy HiGHS | Qiskit Aer

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Minimal Prototype

An end-to-end implemented prototype demonstrating **dynamic quantum-classical blood reallocation** under realistic emergency demand shocks-connecting mathematical formulation to live Qiskit execution.



CORE IMPLEMENTED MODULES (WHAT WE BUILT)

- **Multi-Facility Dataset & Constraints:** Synthetic network modeled on regional blood centres and hospitals, incorporating real-world ABO/Rh inventories and travel-time matrices.
- **Native QUBO & Warm QAOA Solver**: Solves the continuous LP relaxation, biases the Warm QAOA initial state, and samples candidates on Qiskit Aer for independent validation.
- **Bitstring Decoder & Safety Validator:** Classically decodes quantum measurement states and strictly verifies that inventory limits and biological compatibility are never violated.

INTERACTIVE DISRUPTION DEMO (DEMO FLOW)

- **Steady-State Optimization:** Computes baseline allocations that satisfy routine hospital requisitions while minimizing shelf-life expiration risks.
- **Mass-Casualty Incident Shock:** Triggers a real-time surge at a peripheral trauma centre (e.g., O demand jumping 5x with critical urgency).
- **Comparative Decision Support:** After a shock, the system re-optimizes the schedule to divert units to trauma bays and compares candidates with LP, MILP, Greedy and Exact baselines.

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Paired LP warm-start study

METHOD	Mean feasible-shot rate	Paired change	Exact optimum found	Study setting
Standard QAOA	**17.7% mean**	–	**45 / 60** exact hits	**Uniform**
LP warm-start QAOA	**32.1%** 6-case mean	**+14.4 pp** paired mean	**60 / 60** exact hits	LP relaxation warm start
Exact reference	**N/A** reference only	– not compared	**6 / 6** cases solved	Exact reference

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Roadmap

Stage 1: Current MVP

- Synthetic network LP
- Relaxation + QUBO
- Warm QAOA
- simulation Emergency
- simulation LP/MILP/QAOA comparison

Stage 2- Expansion

• Increase number of blood banks and hospitals.
• Study QUBO growth and variable count.
• Introduce repeated dynamic re-optimization.
• Improve visualization and scenario generation.

Stage 3: Advanced quantum research

- Compare Warm QAOA depths(p=1, 2, 3...)
- Ideal vs noisy simulation
- Real quantum hardware, if feasible
- Advanced Warm QAOA (warm-start, constraint-aware mixers)
- Multi-objective trade-offs

Stage 4: Long-term vision

- Real-time inventory and demand updates
- Traffic and travel information
- Dynamic quantum/classical optimization
- Decision-support dashboard
- Future vision, not a deployed clinical system

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References and Sources

- **Quantum algorithm:**

Farhi, E., Goldstone, J., & Gutmann, S. (2014). A Quantum Approximate Optimization Algorithm. arXiv:1411.4028 | Egger et al. (2021), Warm-starting quantum optimization, Quantum 5, 479.

- **Documentation and tools:**

IBM Quantum documentation (QAOA and combinatorial optimization): quantum.cloud.ibm.com/docs
Qiskit documentation: qiskit.org

- **Blood-system context:**

World Health Organization resources on blood safety, availability, emergency blood supply and transfusion compatibility: who.int

- **Our work:**

All code, datasets (synthetic) and experimental results are the team's own implementation. Sources were used for theory, context and methodology.